

Chapter 6

Production Costs

"The art of management is knowing what to overlook."

— William James, *The Principles of Psychology*, 1890

Connecting Week 5 to Week 6: The Analogy Is Now Complete

Consumer Theory (Weeks 1–4)

Optimum = tangency of *indifference curve* + *budget line*.

Condition: $MRS = P_x / P_y$.

Budget line slope = $-P_x / P_y$ (goods prices).

Consumer buys the cheapest bundle that reaches U^* .

Production Theory (Weeks 5–6)

Optimum = tangency of *isoquant* + *isocost line*.

Condition: $MRTS = w / v$.

Isocost slope = $-w / v$ (input prices).

Firm buys the cheapest inputs that produce q^* .

This week delivers the payoff of five weeks of build-up: where costs come from, what shape they take, and why the long run always beats the short run.

Questions This Chapter Will Answer

- Why does an economist say a business earning \$110,000 in accounting profit might still be "losing money"?
- When wages rise, should a firm use more or less capital — and by exactly how much?
- Why does the marginal cost curve always cross the average cost curve at its lowest point?
- Why can a firm in the short run be stuck using a suboptimal input mix even when it knows the optimal one?
- What does it mean for a firm to operate at its "efficient scale"?
- Why does the long-run average cost curve lie below every short-run average cost curve?

Learning Objectives and Topics

Each objective states what you will be able to *do*.

LO1 Distinguish economic from accounting cost; identify opportunity and sunk costs.

§6.1 *Basic Cost Concepts*

LO2 Draw and interpret the isocost line; explain how input-price changes shift it.

§6.2 *Total Costs and the Isocost Line*

LO3 Apply the cost-minimisation condition $MRTS = w/v$ to find optimal input demands.

§6.3 *Profit Max and Cost Min Approach* §6.4 *Graphical Approach*

LO4 Derive the total cost function; relate its shape

to returns to scale.

§6.5 *Total Cost Function*

LO5 Compute AC and MC; prove that MC crosses AC at AC's minimum.

§6.6 *Average and Marginal Costs*

LO6 Distinguish economies of scale from returns to scale; find the efficient scale.

§6.7 *Economies of Scale*

LO7 Derive all short-run cost curves; explain why $STC \geq TC_{LR}$ for all q .

§6.8 *Short Run Envelope* §6.9 *Long Run* §6.10 *LRAC*

Notation Used in This Chapter.

w	wage rate (cost/unit labour)	$TC(q)$	total cost function
v	rental rate of capital	AC	average cost = TC/q
$TC = wL + vK$	total cost identity	MC	marginal cost = dTC/dq
π	economic profit	TFC	total fixed cost
L^*, K^*	cost-minimising inputs	TVC	total variable cost
STC	short-run total cost	$ATC = AFC + AVC$	average total cost
SAC, SMC	short-run AC and MC	AFC	average fixed cost

Before You Begin: Quick Self-Assessment

Test yourself on these prerequisites from Chapters 1–5 before reading:

1. **What is opportunity cost?** *The value of the best alternative foregone when a decision is made — not just cash paid.*
2. **What does the isoquant represent, and what is its slope?** *All (L, K) combinations producing the same output q ; slope = $-MRTS = -MP_L/MP_K$.*
3. **State the cost-minimisation condition from Chapter 5.** $MP_L/MP_K = w/v$, or equivalently $MP_L/w = MP_K/v$ (equal bang-for-the-buck).
4. **What are increasing returns to scale? Give an example.** *Doubling all inputs more than doubles output. Example: a Cobb-Douglas $q = K^{0.6}L^{0.6}$ has IRS since $0.6 + 0.6 > 1$.*
5. **What is the difference between a sunk cost and a fixed cost?** *A sunk cost is already paid and irrecoverable; a fixed cost is owed regardless of output but may be avoidable in the long run (e.g. by exiting).*

Sections 6.1–6.4 cover opportunity cost, isocost, and cost minimisation. Sections 6.5–6.10 build the full cost-curve toolkit.

Chapter Road Map. This chapter converts the production framework of Chapter 5 into a complete cost toolkit. §6.1 distinguishes economic from accounting cost and disposes of sunk costs. §6.2–6.4 build the isocost line and solve the cost-minimisation problem graphically and algebraically. §6.5 derives the total cost function by repeating that optimisation across all output levels. §6.6 introduces average and marginal cost and proves that MC crosses AC at AC's minimum. §6.7 defines economies of scale and the efficient scale. §6.8–6.10 contrast short-run and long-run cost curves and show that the LRAC is the lower envelope of all short-run average cost curves. Chapter 7 uses this entire toolkit to determine the profit-maximising output.

6.1. Basic Cost Concepts

Maria quit her \$45,000 job, earns \$60,000 running a coffee cart, and pays \$8,000 in supplies. Her accountant says she made \$52,000 profit. Her economist friend says she made \$6,500. Who is right — and why does the gap matter?

Mini Case Study: Maria's Coffee Cart. Maria quit her \$45,000-per-year office job to run a coffee cart. She invested \$10,000 of her own savings and pays \$8,000/year in supplies and

permits. Revenue: \$60,000.

Accountant: $\$60,000 - \$8,000 = \$52,000$ profit.

Economist: Add foregone salary (\$45,000) and foregone interest on savings (\$500). Economic profit = $\$60,000 - \$8,000 - \$45,000 - \$500 = +\$6,500$.

Maria is still profitable, but the true surplus above all alternatives is \$45,500 smaller than the accounting figure suggests.

6.1.1. Opportunity Cost, Accounting Cost, Economic Cost

Intuition Before the Definitions. Whenever a firm uses a resource, it gives up whatever else that resource could have earned. That foregone value is the *opportunity cost* — and it is always relevant to decisions, even when no money changes hands. The economist's cost concept captures this; the accountant's does not.

Opportunity Cost: Value of the best alternative foregone when a decision is made.

Accounting Cost: Actual cash payments — what appears on financial statements (explicit costs only).

Economic Cost: Accounting cost **plus** all opportunity costs. Does **not** include sunk costs.

Economic Profit: Total revenue — economic cost. Can be negative even when accounting profit is positive.

In words: Economic cost = every sacrifice the firm makes to produce, whether or not a cheque is written.

*Economic profit
< 0: resources earn
more in their next
best use.*

6.1.2. Sunk Costs: Irrelevant to Forward-Looking Decisions

Sunk Cost: A cost already incurred and unrecoverable. Sunk costs are **irrelevant** to forward-looking choices — they do not affect marginal costs and should never drive the decision to continue or stop.

In words: Sunk costs are gone. Never let a sunk cost change a forward-looking decision.

Example: The \$80 Million Film. A studio has spent \$80m producing a film. Revenue forecast: only \$50m. A \$10m marketing campaign would boost revenue to \$70m. The \$80m is **sunk** — ignore it.

- Release without campaign: net \$50m.
- Run campaign: net $\$70m - \$10m = \$60m$.
- Shelf: net \$0.

Run the campaign ($\$60m > \$50m > \$0$). The \$80m does not enter the decision.

Sunk Cost Fallacy: “We’ve spent so much, we can’t quit now.” This is a systematic decision error — letting an irrecoverable past expenditure drive a current choice. The relevant question is always: going forward, do expected revenues exceed expected costs?

Why This Matters. A small business owner using her own building pays zero rent to herself, so accounting cost is zero. But if the building could fetch \$35,000/year in rent, and the business earns only \$30,000 in accounting profit, it is making an economic *loss* of \$5,000. Understanding economic vs. accounting profit is essential for evaluating whether a business truly creates value above all alternatives.

Check Your Understanding

1. A freelancer earns \$80,000; explicit costs \$12,000; turned down a \$95,000 job offer. Accounting profit? Economic profit? Is she making a rational choice?
2. A firm paid \$500,000 for a machine that is now worthless on the resale market. Is this a sunk cost? How should it affect the firm's current production decisions?
3. True or false: "A positive accounting profit always means the firm is making good use of its resources."

We know how economists measure costs. The next step is to express those costs as a budget for the firm's input purchases — the isocost line.

6.2. Total Costs and the Isocost Line

A firm has a cost budget of \$60 and faces wage \$6 and capital rental rate \$3. What are all the ways it can spend exactly \$60 on labour and capital? And what happens to those options when the wage rises?

6.2.1. Total Cost Identity

With labour L at wage w and capital K at rental rate v :

$$TC = wL + vK \quad (6.1)$$

In words: Total spending = wage bill + capital bill.

6.2.2. The Isocost Line

Intuition Before the Formula. The isocost line is to the firm exactly what the budget line is to the consumer. The budget line showed all affordable consumption bundles at a given income; the isocost shows all input combinations with the same total cost. Same mathematics, same graphical logic — the only difference is that we now minimise cost rather than maximise utility.

Isocost Line: All (L, K) combinations with total cost equal to C :

$$C = wL + vK \implies K = \frac{C}{v} - \frac{w}{v}L \quad (6.2)$$

Straight line: K -intercept $= C/v$, L -intercept $= C/w$, slope $= -w/v$.

In words: Slope $-w/v$: to hire one more unit of labour, the firm must give up w/v units of capital to keep

spending constant.

How changes affect the isocost:

- C increases → **parallel outward shift** (both intercepts rise, slope unchanged).
- w rises → **rotation inward around K -intercept** (L -intercept falls, K -intercept fixed).
- v rises → **rotation inward around L -intercept** (K -intercept falls, L -intercept fixed).

Isocost slope
 $= -w/v$. Mirrors
 budget line slope
 $= -P_x/P_y$.

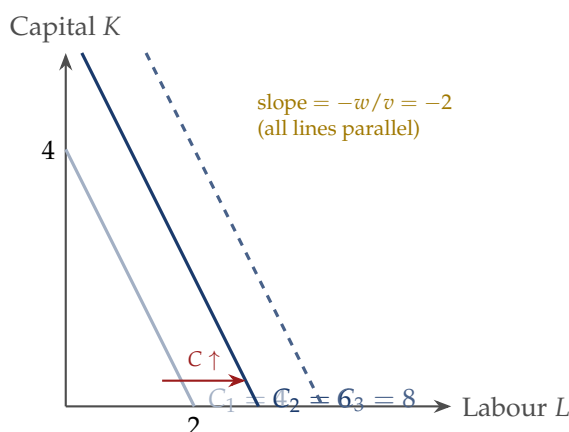


Figure 6.1. Isocost line family with slope $= -w/v = -2$. What to look for: All three lines are parallel — the slope depends only on w/v , not on C . Moving outward represents a higher cost budget. A change in w would rotate each line; a change in C alone gives a parallel shift.

Application: Minimum Wage and Capital Substitution. When a government raises the minimum wage, w rises while v is unchanged. The isocost line *rotates inward* around the K -intercept. For a firm previously at its cost minimum, MP_L/w now falls (labour is more expensive). The bang-for-the-buck condition breaks: $MP_L/w < MP_K/v$. The firm shifts spending toward capital and away from labour — moving along its isoquant to a more capital-intensive combination. This is the economic logic behind automation following minimum wage increases: ATMs replacing bank tellers, self-checkout kiosks replacing cashiers, robots replacing assembly workers.

Check Your Understanding

1. $w = \$10$, $v = \$5$, $C = \$100$. Write the isocost, find both intercepts, state the slope.
2. The wage doubles to $\$20$ (all else unchanged). Which intercept is fixed? Which moves? What happens to the slope?
3. Both w and v double. Does the isocost slope change? Do the intercepts change? What happens to the cost-minimising K/L ratio?

The isocost describes equal-cost combinations. Combining it with the isoquant lets us find the cheapest way to produce a given output.

6.3. Profit Maximisation and Cost Minimisation

A firm wants to maximise profit. Why is this equivalent to minimising costs for a given output — and why does that equivalence simplify the whole problem?

Economic Profit:

$$\pi = \underbrace{P \cdot f(K, L)}_{\text{Total Revenue}} - \underbrace{(wL + vK)}_{\text{Total Cost}} \quad (6.3)$$

In words: Profit = revenue minus all economic costs. Negative even when accounting profit is positive.

When output target q_1 is fixed, revenue $P \cdot q_1$ is constant. Maximising π reduces to minimising cost:

Cost Minimisation Problem:

$$\min_{L, K} wL + vK \quad \text{subject to} \quad f(K, L) = q_1 \quad (6.4)$$

Profit max $\Leftrightarrow$ cost min (for given output). Solving the cost problem solves the profit problem.

Two Problems, One Solution. This equivalence lets us split a complex problem — how much to produce *and* how to produce it — into two clean steps: (1) find the cheapest way to produce any given quantity (Week 6); (2) choose which quantity to produce (Week 7 onwards).

Check Your Understanding

1. Revenue \$500, economic costs \$480: economic profit? If accounting costs were \$420, accounting profit? Is the firm making a rational economic choice?
2. “Minimising cost is always the right objective.” True or false? When is it precisely correct?
3. Why does fixing output convert the profit problem into a pure cost problem?

Cost minimisation has been set up formally. Now we solve it graphically.

6.4. Cost Minimisation: Graphical Approach

The firm wants to produce q_1 at minimum cost. Graphically this is like the consumer finding the highest indifference curve on a budget — except now we want the lowest isocost touching the isoquant. What does the solution look like, and what is the optimality condition?

6.4.1. The Tangency Condition

Intuition. The cost minimum is the lowest isocost line still touching the isoquant for q_1 . Any isocost below it is unattainable; any isocost above it is feasible but wasteful. The minimum occurs at the tangency point, where the slopes of the isoquant and isocost are equal.

Setting the slopes equal gives the **cost-minimisation condition**:

$$\frac{MP_L}{MP_K} = \frac{w}{v} \quad (6.5)$$

In words: The MRTS must equal the input price ratio. The firm substitutes labour for capital in production at the same rate as the market substitutes them in cost.

Rearranging into the “**bang for the buck**” form:

$$\frac{MP_L}{w} = \frac{MP_K}{v} \quad (6.6)$$

MRTS = w/v is the production mirror of the consumer's $MRS = P_x/P_y$.

In words: At the optimum: output per dollar spent on labour = output per dollar spent on capital. If one side is larger, shift spending toward that input.

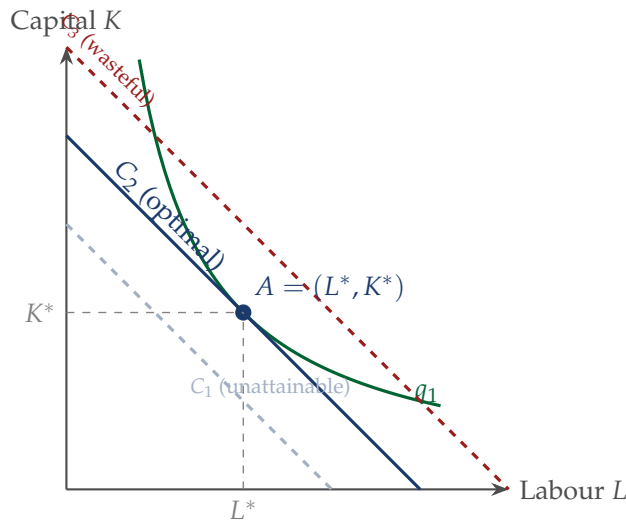


Figure 6.2. Cost minimisation: tangency at A. What to look for: C_1 is unattainable. C_3 is feasible but wasteful. C_2 is optimal: the lowest isocost touching the isoquant, at tangency point A where $MP_L/MP_K = w/v$. Exactly mirrors the consumer's optimum where $MRS = P_x/P_y$.

Worked Example: Checking the Optimality Condition.

$q = \sqrt{KL}$, $w = 4$, $v = 1$. Is $(L, K) = (4, 16)$ cost-minimising for $q = 8$?

Step 1 — Marginal products:

$$MP_L = \frac{1}{2}\sqrt{K/L} = \frac{1}{2}\sqrt{16/4} = 1; \quad MP_K = \frac{1}{2}\sqrt{L/K} = \frac{1}{2}\sqrt{4/16} = \frac{1}{4}$$

Step 2 — Check condition (Eq. 6.5):

$$\frac{MP_L}{MP_K} = \frac{1}{1/4} = 4 = \frac{w}{v} = \frac{4}{1} \checkmark$$

Step 3 — Output: $q = \sqrt{16 \times 4} = 8$. $\checkmark$

Step 4 — Cost: $TC = 4(4) + 1(16) = \$32$.

In words: At $(4, 16)$: the firm is cost-minimising for $q = 8$ at total cost \$32.

Economic Insight. The cost-minimisation condition $MP_L/MP_K = w/v$ is the production mirror of the consumer's $MRS = P_x/P_y$. Both are tangency conditions: the subjective trade-off rate (MRS or MRTS) equals the market trade-off rate (price ratio). This structural identity is the payoff of the five-week analogy thread.

Common Mistake: Wrong reallocation direction. When $MP_L/w > MP_K/v$, shift spending toward labour, away from capital. Students sometimes increase both inputs instead. Cost minimisation is about reallocation *across* inputs, not adding more of everything.

Check Your Understanding

1. $MP_L = 40$, $MP_K = 20$, $w = \$8$, $v = \$5$. Is the firm cost-minimising? If not, which input should it use more of?
2. For $q = K^{1/3}L^{2/3}$: derive MP_L and MP_K , apply Eq. 6.5, and express K/L as a function of w and v .
3. “At the cost minimum, both inputs have equal marginal products.” True or false?

In Chapter 5 we found the cost-minimising input combination for each output level q ; the total cost of that combination at each q is exactly the cost function $TC(q)$ that we now analyse. The optimality condition gives the best input mix for any output level; applying it across all output levels traces out the firm's total cost function.

6.5. The Total Cost Function

We know how to find the cheapest input combination for one specific output level. What happens when we repeat this for every possible output level and connect the dots?

6.5.1. Deriving the Total Cost Function

Intuition. The total cost function $TC(q)$ is the firm's “menu of minimum costs” — it tells the firm the cheapest possible cost for producing any output q , given input prices w and v .

Procedure:

Step 1. Pick output level q_1 .

Step 2. Apply $MP_L/MP_K = w/v$ and $f(L, K) = q_1$ to find cost-minimising (L_1^*, K_1^*) .

Step 3. Compute $TC(q_1) = wL_1^* + vK_1^*$.

Step 4. Repeat for all q . The resulting $TC(q)$ curve is the total cost function.

6.5.2. Shape of the Total Cost Curve

The shape depends on the underlying returns to scale:

TC Shape and Returns to Scale

Technology	TC shape	Why
CRS	Straight line through origin	Doubling q doubles inputs and costs
IRS	Concave (flattens)	Doubling q needs less than double inputs
DRS	Convex (steepens)	Doubling q needs more than double inputs
Typical firm	Inverse-S (concave then convex)	IRS at low q ; DRS at high q

Common Mistake: TC always starts at the origin. In the *long run* with no fixed inputs, $TC(0) = 0$ and the curve can start at the origin. In the *short run*, fixed costs must be paid even at zero output, so the short-run TC curve has a positive y -intercept equal to total fixed cost TFC.

Check Your Understanding

1. A firm with CRS doubles all inputs. What happens to output? To total cost? What shape does its TC curve have?
2. Sketch TC curves for CRS, IRS, and DRS on the same diagram. Which is steepest at high output levels?
3. For $q = \sqrt{KL}$ with $w = 4$, $v = 1$: we showed $TC = 2q\sqrt{wv} = 4q$ (see worked example in Section 6.10). What type of returns to scale does this imply?

The total cost function tells us the overall cost level. To understand cost per unit and the cost of the next unit, we need average and marginal cost.

6.6. Average and Marginal Costs

If total cost is \$300 when producing 10 units and \$315 when producing 11 units, the marginal cost of the 11th unit is \$15. But the average cost is \$30. Why is marginal cost so much lower than average cost here — and what does that tell us about where the firm is on its cost curve?

6.6.1. Definitions

$$AC = \frac{TC}{q} \quad (AC)$$

$$MC = \frac{dTC}{dq} \quad (MC)$$

AC is cost per unit. MC is the additional cost of the next unit — the slope of the TC curve.

In words: AC asks: on average, how much does each unit cost? MC asks: how much does one more unit cost?

“Your average cost tells you how you’ve been doing. Your marginal cost tells you what to do next.”
In economics, it is always the margin that drives decisions — AC is a report card, MC is the compass.

6.6.2. The MC–AC Relationship: Why MC Crosses AC at the Bottom

The Grade Analogy. Suppose a class of 30 students has an average grade of 80. If the 31st student (the “marginal” student) scores 60, the average falls. If she scores 95, the average rises. If she scores exactly 80, the average is unchanged.

This is exactly the MC–AC relationship:

- $MC < AC$: each new unit costs less than average $\Rightarrow$ AC is **falling**.
- $MC > AC$: each new unit costs more than average $\Rightarrow$ AC is **rising**.
- $MC = AC$: the new unit costs exactly the average $\Rightarrow$ AC is at its **minimum**.

MC always crosses AC at AC’s minimum. This is a mathematical certainty.

In words: When the cost of the next unit is below the current average, the average falls. When it is above, the average rises. They meet at the bottom.

Formal Proof: Why MC Crosses AC at AC’s Minimum.

Since $AC = TC/q$, differentiate with respect to q using the quotient rule:

$$\frac{dAC}{dq} = \frac{q \cdot \frac{dTC}{dq} - TC}{q^2} = \frac{q \cdot MC - q \cdot AC}{q^2} = \frac{MC - AC}{q}$$

Therefore:

$$\boxed{\frac{dAC}{dq} = \frac{MC - AC}{q}}$$

- $MC < AC \Rightarrow dAC/dq < 0$: AC is **falling**.
- $MC > AC \Rightarrow dAC/dq > 0$: AC is **rising**.
- $MC = AC \Rightarrow dAC/dq = 0$: AC is at its **minimum**.

The grade analogy is exact: the crossing point is a theorem, not a coincidence.

Why This Matters: MC and Pricing Strategy. MC is the single most important cost concept for pricing. Competitive firms set price equal to MC. Firms with market power set price above MC but still use MC to decide how many units to produce. The U-shaped MC curve reflects a real feature of most production: there is a range where additional units cost less (specialisation, fixed costs spread), then a range where they cost more (diminishing returns, congestion). Knowing where the firm sits on its MC curve is essential for sound pricing.

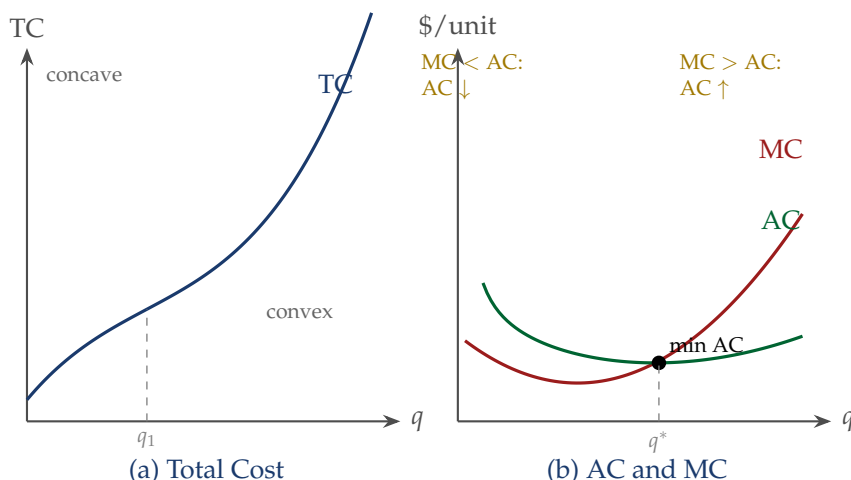


Figure 6.3. TC, AC, and MC curves for a typical firm. *What to look for:* In panel (a), TC is first concave (IRS region) then convex (DRS region). In panel (b), MC is U-shaped (it is the slope of TC). AC is also U-shaped but flatter. MC crosses AC from below at AC's minimum q^* — this is a mathematical identity, proven above, not a coincidence.

Check Your Understanding

1. $TC = 300 + 10q + 2q^2$. Derive AC and MC. At what output does $MC = AC$? What is AC at that point?
2. True or false: "If MC is rising, AC must also be rising." Explain with an example.
3. A class of 20 students has average grade 75. If the 21st student scores 60, does the average rise or fall? Relate this to the MC–AC relationship.

Average and marginal costs tell us about cost per unit as output changes. The next concept — economies of scale — uses this to classify production technologies and explain industry structure.

6.7. Economies of Scale

An airline that doubles its passenger capacity may be able to reduce its cost per passenger. But a craft brewery that doubles its batch size might find costs per bottle rising. What determines which way average cost moves as output expands?

Economies of Scale: AC is **falling** as output increases.

Diseconomies of Scale: AC is **rising** as output increases.

Constant (no) Economies of Scale: AC is **flat** as output increases.

Efficient Scale: The output level q^* that minimises average cost.

In words: Economies of scale: the more you produce, the cheaper each unit gets. Diseconomies: the more you produce, the more expensive each unit gets.

6.7.1. Economies of Scale vs. Returns to Scale

Students frequently confuse these. They are related but distinct:

Returns to Scale vs. Economies of Scale

Returns to Scale	Economies of Scale
Property of the <i>production function</i>	Property of the <i>cost function</i>
Describes output change when all inputs scale	Describes AC change as output increases
Long-run concept (all inputs variable)	Can apply in short run too
IRS $\Rightarrow$ Economies of scale (LR)	Economies of scale may occur even under CRS (SR: fixed costs spread)
DRS $\Rightarrow$ Diseconomies of scale (LR)	
CRS $\Rightarrow$ Flat AC (LR)	

Example: Identifying Economies of Scale.

Firm's AC: $AC(q) = 100/q + q$. Economies of scale at $q = 5$? At $q = 15$?

$$dAC/dq = -100/q^2 + 1.$$

At $q = 5$: $dAC/dq = -100/25 + 1 = -3 < 0 \Rightarrow$ AC falling $\Rightarrow$ **economies of scale**.

At $q = 15$: $dAC/dq = -100/225 + 1 \approx +0.56 > 0 \Rightarrow$ AC rising $\Rightarrow$ **diseconomies of scale**.

Efficient scale: set $dAC/dq = 0$: $q^* = 10$. At $q^* = 10$: $AC = 100/10 + 10 = 20$ (minimum AC).

Application: Economies of Scale and Industry Structure.

Airlines (initial economies, then diseconomies): Larger airlines spread fixed costs (maintenance facilities, reservation systems, gates) over more passengers, reducing AC. Beyond a certain scale, coordination costs and congestion cause diseconomies. The U-shaped LRAC reflects both phases.

Technology platforms (strong economies): Once software is written, serving an extra user costs nearly zero. AC falls steeply as output grows. Natural tendency toward monopoly or oligopoly — a key reason regulators scrutinise technology mergers so closely.

Artisanal production (diseconomies from the start): Quality depends on the master craftsman's personal attention. AC rises with output because the binding constraint is skilled human time that cannot be replicated at scale.

Going deeper: Industries where LRAC falls steeply throughout (utilities, railroads, broadband networks) are *natural monopolies*: one large firm can always undercut any smaller competitor. This is why regulators either allow a regulated monopoly or require structural separation.

The Economist's Lens: Hospital Mergers and Economies of Scale

The concept in practice: Why hospitals merge — and whether it works.

Hospital mergers have reshaped US healthcare over the past three decades, with the number of community hospitals falling by over 20% since 1990 even as total hospital beds held roughly steady. The economic logic offered by every merger prospectus is the same: economies of scale. Larger hospitals can spread fixed costs — expensive imaging equipment, specialist staff, electronic health record systems, administrative overhead — over more patients, reducing average cost per admission.

Does the evidence support this? Empirical studies of hospital cost functions consistently find that economies of scale are **exhausted at relatively modest size**. Dranove (1998) and subsequent work estimate that the minimum efficient scale for most hospital services corresponds to roughly 200–250 beds, with average cost declining up to that point and then becoming approximately flat or rising thereafter. Estimates from the empirical literature suggest that output elasticities for hospital inputs are close to 1.0 at large scale, implying $a + b \approx 1$ — consistent with CRS rather than strong IRS — for hospitals well above 300 beds.

This creates a striking tension: mergers are overwhelmingly driven by a cost rationale, yet the academic evidence suggests the cost savings from scale are largely exhausted before hospitals reach the sizes at which most mergers occur. The Federal Trade Commission has cited this literature in challenging hospital mergers that it believes would raise prices without generating the claimed cost efficiencies. The real beneficiary of many hospital mergers, several studies argue, is not reduced AC but increased bargaining power with insurers — a market power story, not a cost story.

What this illustrates: The LRAC curve is a concrete, empirically estimable object. Regulators and courts use cost-function estimates to evaluate whether a proposed merger is likely to generate genuine efficiency gains (moving down the LRAC) or merely to concentrate market power. Economies of scale is not just classroom theory — it is a central exhibit in competition law.

Source: Dranove (1998, *RAND Journal of Economics*); Dafny (2009, *AER*); Federal Trade Commission hospital merger guidelines (2023).

Check Your Understanding

1. A firm has CRS in production. In the long run, is its LRAC curve flat, U-shaped, or something else? What about in the short run when it has fixed costs?
2. For $AC = q^2 - 8q + 20$: find the efficient scale. What is the minimum AC?
3. Why do industries with strong economies of scale tend to have fewer, larger firms than industries with diseconomies?

We have now covered costs in the long run where all inputs are variable. The next sections introduce the short run, where capital is fixed, and show why short-run costs always exceed long-run costs.

6.8. The Short Run and the Long Run

An airline can hire extra cabin crew within days but needs three years to commission new aircraft. A wheat farmer plants in spring and harvests in autumn — during that period, land is fixed. What determines the

length of the “short run,” and how does a fixed input change the firm’s cost structure?

6.8.1. Definitions

Short Run: At least one input is **fixed** — the firm cannot change its quantity regardless of desired output. In this course: capital $\bar{K}$ is fixed; only labour L is variable.

Long Run: All inputs are **variable** — the firm can freely choose any (L, K) combination.

The Short Run Is About Flexibility, Not a Fixed Time Period. The short run is not “one year.” It is the period during which at least one input cannot be adjusted. For a nuclear power plant, the short run (before new capacity is possible) may be decades. The defining characteristic is the *inability to vary a factor*, not the calendar length.

“A sunk cost is a sunk cost. A fixed cost is a fixed cost. The difference: you already paid the sunk cost; you are about to pay the fixed cost. Neither should change what you produce next — but only one of them you can avoid by shutting down.” Fixed costs matter for long-run exit; sunk costs never matter.

6.8.2. Short-Run Cost Decomposition

With capital fixed at $\bar{K}$:

$$\text{TFC} = v\bar{K} = \text{constant (paid regardless of output)} \quad (6.7)$$

$$\text{TVC} = wL(q) \quad (\text{rises with output}) \quad (6.8)$$

$$\text{STC} = \text{TFC} + \text{TVC} = v\bar{K} + wL(q) \quad (6.9)$$

Derived cost measures:

$$\text{ATC} = \frac{\text{STC}}{q} = \text{AFC} + \text{AVC}; \quad \text{AFC} = \frac{\text{TFC}}{q} = \frac{v\bar{K}}{q}; \quad \text{AVC} = \frac{\text{TVC}}{q}$$

$$\text{SMC} = \frac{d\text{STC}}{dq} = \frac{d\text{TVC}}{dq} \quad (\text{fixed costs do not affect MC})$$

In words: SMC equals the rate of change of variable costs only. Fixed costs never affect how much the next unit costs to produce.

Fixed Costs Are Irrelevant to Marginal Decisions. Because TFC is constant, $d\text{TFC}/dq = 0$, so $\text{SMC} = d\text{TVC}/dq$. A firm deciding whether to produce 100 or 101 units should base that decision on variable costs only. Fixed costs are “water under the bridge” for short-run production decisions. (But they matter for long-run entry/exit decisions.)

6.8.3. Properties of Short-Run Cost Curves

Five key properties:

1. $\text{ATC} = \text{AFC} + \text{AVC}$ at every output level.

2. AFC is always *falling* (fixed cost spread over more units).
3. SMC passes through the minimum of both AVC and ATC.
4. The minimum of ATC occurs to the *right* of the minimum of AVC (because falling AFC continues to pull ATC down even after AVC starts rising).
5. SMC is U-shaped; ATC and AVC are U-shaped; AFC is always decreasing.

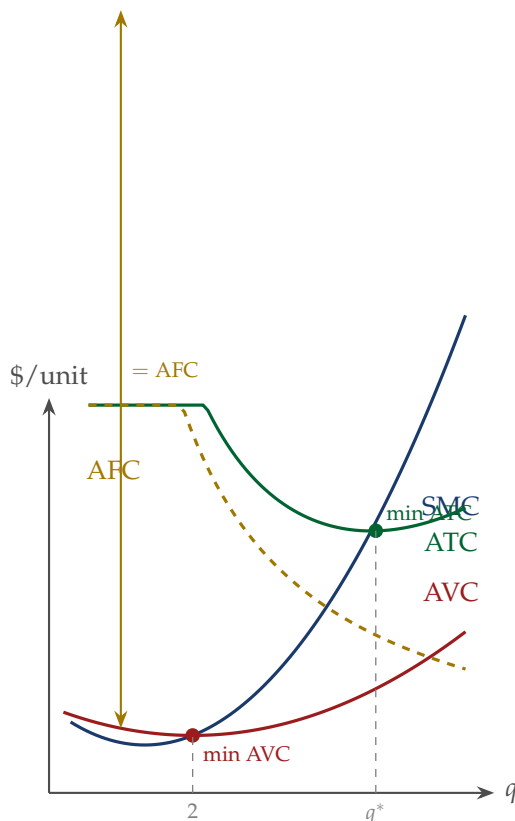


Figure 6.4. Short-run cost curves. *What to look for:* SMC (blue) is U-shaped and passes through the minimum of both AVC (red, at $q = 2$) and ATC (green, at $q \approx 4.6$). AFC (gold, dashed) falls continuously — the vertical gap between ATC and AVC equals AFC at every output. Minimum ATC lies to the *right* of minimum AVC because falling AFC continues pulling ATC down after AVC starts rising. Parameterisation: $STC = 50 + 6q - 2q^2 + 0.5q^3$; all values algebraically verified.

6.8.4. The Short Run Constrains the Firm

In the long run, the firm can move along an isoquant to its optimal point. In the short run, capital is fixed at $\bar{K}$ — the firm is stuck at the point where the horizontal line $K = \bar{K}$ meets the isoquant, which is generally not the long-run optimum.

Short-run costs $\geq$ Long-run costs at every output level.

In the long run, the firm has complete flexibility to choose the cheapest input combination. In the short run, it is constrained to one level of capital. The constraint can only raise costs relative to the unconstrained optimum. Equality holds only at the output level for which $\bar{K}$ happens to be exactly the long-run optimal capital.

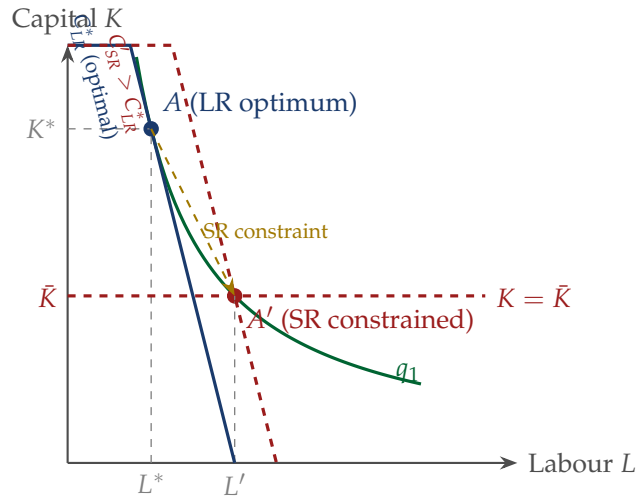


Figure 6.5. The short-run constraint. What to look for: A is the long-run cost-minimising point ($MRTS = w/v$ satisfied). With capital fixed at $\bar{K} < K^*$, the firm is forced to point A' — more labour, less capital than optimal, on a higher isocost $C'_{SR} > C^*_{LR}$. The short-run constraint raises costs above the long-run minimum at this output level. Equality holds only where $\bar{K} = K^*_{LR}(q)$.

Check Your Understanding

1. $STC = 500 + 4q + 0.5q^2$. Identify TFC and TVC. Derive AFC, AVC, ATC, SMC. At what output does $SMC = AVC$?
2. Why does a firm in the short run sometimes operate at a point where $MRTS \neq w/v$? Use Figure 6.5 to explain.
3. True or false: “A firm should shut down in the short run whenever it makes an accounting loss.” Explain using fixed costs.

We have seen how short-run costs differ from long-run costs at any fixed capital level. The final step is to show how all these short-run cost curves combine to trace out the long-run average cost curve.

6.9. The Long Run: LRAC as the Envelope of SRAC Curves

Different plant sizes correspond to different short-run cost curves. In the long run, the firm can choose its plant size. How does it combine all these short-run curves into a single long-run average cost curve?

Intuition. Think of each SRAC curve as representing a different “version” of the firm with a particular plant size. In the long run, the firm selects the plant size that minimises average cost at each output level. The LRAC curve is the *lower envelope* — the lowest available SRAC at every output.

Key properties of the LRAC curve:

- $LRAC \leq SRAC$ at every output (the long run cannot be more expensive than the short run).

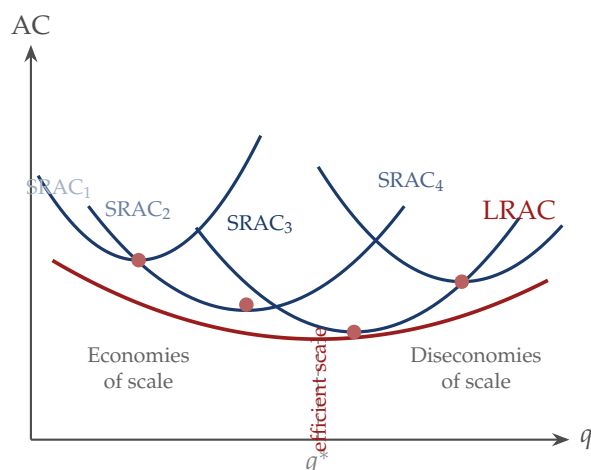
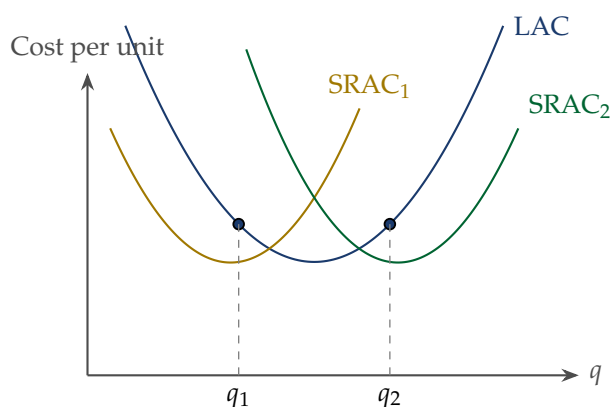


Figure 6.6. LRAC is the lower envelope of SRAC curves. *What to look for:* Each SRAC corresponds to a fixed plant size. In the long run, the firm selects the plant minimising cost at each output. The LRAC (red) is tangent to each SRAC from above. It is U-shaped for a typical firm: falling (economies of scale) until the efficient scale q^* , then rising (diseconomies). $LRAC \leq SRAC$ at every output level.

- LRAC is *tangent* to each SRAC curve from above.
- LRAC is typically **U-shaped**: economies of scale at low q , efficient scale at the minimum, diseconomies at high q .
- The LRAC is **flatter and smoother** than any individual SRAC curve.
- Equality $LRAC = SRAC$ holds only at the output where the firm's current capital is exactly the long-run optimal level.



What this shows: Each SRAC curve is tangent to the LRAC at the output for which the plant was designed (q_1 for $SRAC_1$, q_2 for $SRAC_2$). At any other output, $SRAC > LRAC$ — short-run costs always equal or exceed long-run costs, with equality only at the tangency point.

Why This Matters: Plant Size and Long-Run Strategy. The LRAC is a roadmap for long-run business strategy. Firms to the left of q^* can reduce unit costs by expanding. Firms to the right face diseconomies and may benefit from downsizing. Industries where LRAC falls steeply

(utilities, broadband, cloud computing) tend toward natural monopoly. Industries with flat LRAC sustain many competitors of similar size.

Historical Episode: US Airline Deregulation (1978) and the Shape of the LRAC.

Before 1978, the Civil Aeronautics Board (CAB) regulated US airline routes and fares, shielding carriers from competition and allowing inefficient operations to persist. The Airline Deregulation Act of 1978 removed these protections and exposed carriers to market competition for the first time. The result was a dramatic, decade-long natural experiment in the shape of airline cost curves.

Two findings emerged that directly illustrate LRAC theory. First, **economies of scale**: deregulation revealed that larger carriers operating hub-and-spoke networks — which concentrated passengers through a small number of major airports — had substantially lower cost per passenger-mile than point-to-point carriers of the same size. The hub structure spread fixed costs (gate leases, maintenance facilities, reservation systems) over more passengers. Average costs fell as network size grew. Second, **efficient scale**: the gains from scale were not unlimited. By the mid-1980s, studies by Caves, Christensen, and Tretheway (1984) found that the LRAC for major carriers was roughly L-shaped — steeply declining at small scale, then flat or modestly rising beyond roughly 10 billion revenue passenger-miles per year. Beyond that output, coordination complexity began to offset scale economies.

The competitive shakeout confirmed the theory: carriers that could not reach efficient scale (many regional carriers and new entrants) either exited or were acquired. Those who grew past efficient scale, like early-expansion Pan Am, found that size alone did not guarantee low costs. Deregulation turned the LRAC curve from an abstract concept into a survival constraint.

Policy implication: When markets are regulated, firms may operate far from their efficient scale. Deregulation forces convergence toward q^* — those who adapt survive; those who cannot exit. The LRAC is not just a diagram; it is a selection mechanism.

Source: Caves, Christensen & Tretheway (1984, *RAND Journal of Economics*); Morrison & Winston (1986), *The Economic Effects of Airline Deregulation*.

Check Your Understanding

1. Sketch LRAC and two SRAC curves. Mark: (a) the point where $SRAC_2$ is tangent to LRAC; (b) a point where $SRAC_2$ lies *above* LRAC.
2. Why can the LRAC never lie *above* any SRAC curve? What would it mean economically if LRAC were higher than SRAC at some output?
3. A firm operates on $SRAC_3$ but currently produces less than the output at which $SRAC_3$ is tangent to LRAC. Is it operating at efficient scale? What should it do in the long run?

The theory is complete. The next section brings everything together through two fully worked numerical examples.

6.10. Worked Examples

How do we apply the cost-minimisation condition, derive the total cost function, and compute all cost curves in a single unified example?

6.10.1. Long-Run Cost: $q = \sqrt{KL}$

Example: Deriving Long-Run TC, AC, and MC.

$q = \sqrt{KL}$; input prices w and v .

Step 1 — Marginal products:

$$MP_L = \frac{1}{2}\sqrt{K/L}; \quad MP_K = \frac{1}{2}\sqrt{L/K}$$

Step 2 — Optimality condition (Eq. 6.5):

$$\frac{K}{L} = \frac{w}{v} \implies K = \frac{w}{v}L$$

Step 3 — Solve for L^* and K^* :

$$q = \sqrt{\frac{w}{v}L \cdot L} = L\sqrt{w/v} \implies L^* = q\sqrt{v/w}; \quad K^* = q\sqrt{w/v}$$

Step 4 — Total cost:

$$TC(q) = wL^* + vK^* = w \cdot q\sqrt{v/w} + v \cdot q\sqrt{w/v} = q\sqrt{wv} + q\sqrt{wv} = 2q\sqrt{wv} \quad (6.10)$$

In words: TC is proportional to q : straight-line cost curve, consistent with CRS ($a + b = 0.5 + 0.5 = 1$).

Numerical check ($w = 4, v = 1$): $TC(q) = 2q\sqrt{4} = 4q$. At $q = 8$: $TC = 32$, consistent with the worked example in Section 6.4 at $(L, K) = (4, 16)$. ✓

Step 5 — AC and MC:

$$AC(q) = \frac{2q\sqrt{wv}}{q} = 2\sqrt{wv}; \quad MC(q) = \frac{dTC}{dq} = 2\sqrt{wv}$$

$AC = MC = 2\sqrt{wv}$ at all output levels — consistent with CRS and flat cost curves.

Economic Insight. For CRS, $AC = MC$ everywhere and both are constant. This is the long-run benchmark: neither economies nor diseconomies of scale.

6.10.2. Short-Run Cost: Hockey Stick Manufacturer

Example: Short-Run Costs for $q = 2\sqrt{KL}$, $\bar{K} = 100, w = 4, v = 1$.

Part (a) — Derive STC and SAC.

Given $K = 100$ fixed: $q = 2\sqrt{100 \cdot L} = 20\sqrt{L} \implies L = q^2/400$.

$$STC = v\bar{K} + wL = 100 + 4 \cdot \frac{q^2}{400} = 100 + \frac{q^2}{100} \quad (6.11)$$

$$SAC = \frac{100}{q} + \frac{q}{100} \quad (6.12)$$

Part (b) — Short-run marginal cost:

$$SMC = \frac{dSTC}{dq} = \frac{2q}{100} = \frac{q}{50} \quad (6.13)$$

In words: SMC rises linearly — each extra unit of output requires progressively more labour because of diminishing marginal product.

Part (c) — Numerical table (verified):

q	STC	SAC	SMC	Relationship
25	106.25	4.25	0.50	SMC < SAC: SAC falling
50	125.00	2.50	1.00	SMC < SAC: SAC falling
100	200.00	2.00	2.00	SMC = SAC: minimum SAC
150	325.00	2.17	3.00	SMC > SAC: SAC rising
200	500.00	2.50	4.00	SMC > SAC: SAC rising

Part (d) — Verify minimum SAC analytically: Set $SMC = SAC$: $q/50 = 100/q + q/100 \implies q/50 - q/100 = 100/q \implies q/100 = 100/q \implies q^2 = 10,000 \implies q^* = 100$. ✓

Part (e) — Long-run comparison note: From Section 6.10, $TC_{LR} = 2q\sqrt{wv} = 4q$ (with $w = 4, v = 1$). The SR and LR costs coincide at the unique output where $\bar{K} = K_{LR}^*(q)$. Setting $K_{LR}^* = q\sqrt{w/v} = 2q = 100$ gives $q = 50$. At $q = 50$: this is the output for which $\bar{K} = 100$ is exactly the long-run optimal capital, and $STC = TC_{LR}$. At all other output levels, $STC \geq TC_{LR}$ — the short-run constraint raises costs.

Try It. $STC = 200 + 8q + 2q^2$. (a) Identify TFC and TVC. (b) Derive SMC, AVC, AFC, ATC. (c) At what output does $SMC = AVC$? (d) Verify numerically at $q = 5$: is $ATC = AFC + AVC$?

The worked examples demonstrate the full toolkit. Before turning to the practice questions, let us collect the exam mistakes, revision guide, and supplementary materials.

The cost framework is now complete: from economic vs. accounting cost, through cost minimisation, to the full suite of short-run and long-run cost curves. The practice questions apply all of these concepts — both forward (given the production function, derive cost curves) and in reverse (given observed cost data, recover the underlying technology). Attempt every question before consulting the solutions.

Where Does This Take Us?

- **Week 7** introduces market structure. The condition “price = marginal cost” for competitive firms comes directly from the cost framework here. The short-run supply curve is derived from the SMC curve.
- **The shut-down condition** (short run) and the **exit condition** (long run) both involve comparing price against minimum AVC and minimum LRAC respectively.
- **Natural monopoly** arises when LRAC is falling throughout the relevant output range — the cost framework explains directly why one large firm can always undercut many small ones.
- **Producer surplus** (Week 7) is the area above the supply curve and below price — the mirror of consumer surplus from Week 4. Total surplus = CS + PS gives the welfare benchmark against which all market interventions are measured.

Five Common Exam Mistakes — and How to Avoid Them

1. **Drawing the short-run TC curve starting at the origin.** The short-run TC curve has a positive y -intercept equal to total fixed cost. Only the long-run TC curve (with no fixed inputs) starts at or near the origin. Always ask: am I in the short run or long run before drawing TC.
2. **Confusing returns to scale with economies of scale.** Returns to scale is a property of the production function (what happens to output when all inputs scale). Economies of scale is a property of the cost function (what happens to AC as output rises). A firm with CRS in production still exhibits economies of scale in the short run because fixed costs are spread over more units.
3. **Reallocating in the wrong direction when bang-for-the-buck is unequal.** When $MP_L/w > MP_K/v$, the firm should use *more* labour and *less* capital. It should not simply add more labour without reducing capital — that would change output, not minimise cost for the given output.
4. **Thinking SMC passes through the ATC minimum and AVC minimum at the same point.** SMC intersects AVC at AVC’s minimum, and ATC at ATC’s minimum — but these are two different output levels. The ATC minimum always lies to the *right* of the AVC minimum, because declining AFC continues to pull ATC down after AVC starts rising.
5. **Arguing that SRAC can lie below LRAC somewhere.** LRAC is the lower envelope of all SRAC curves: $SRAC \geq LRAC$ at every output level, with equality only at the tangency point. If your diagram shows a SRAC curve below LRAC, something is wrong — the long run is always at least as good as the short run.

Chapter at a Glance — Week 6: Production Costs

Your one-page revision guide.

Key Equations

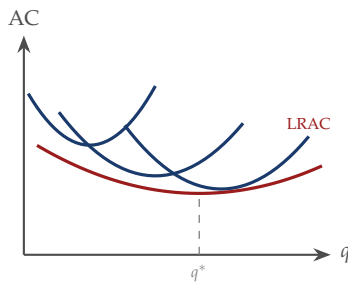
Total cost	$TC = wL + vK$	p.5
Isocost	$K = C/v - (w/v)L$	p.5
Economic profit	$\pi = Pq - TC$	p.7
Cost min problem	$\min wL + vK \text{ s.t. } f = q$	p.7
Optimality cond.	$MP_L/MP_K = w/v$	p.8
Bang for the buck	$MP_L/w = MP_K/v$	p.8
Short-run TC	$STC = TFC + TVC$	p.15
MC-AC identity	$dAC/dq = (MC - AC)/q$	p.10

Economies of Scale Summary

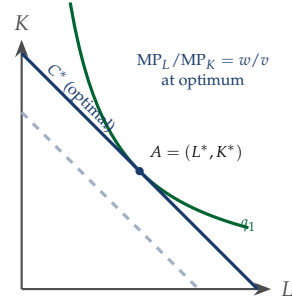
IRS in production	$\Rightarrow$	AC falling (LR)
DRS in production	$\Rightarrow$	AC rising (LR)
CRS in production	$\Rightarrow$	AC constant (LR)

SR economies of scale possible even under CRS

LRAC Envelope



Cost Minimisation Diagram



SR Cost Curve Properties

- $ATC = AFC + AVC$ always
- AFC always falling
- SMC passes through $\min(AVC)$ and $\min(ATC)$
- $\min(ATC)$ is to the *right* of $\min(AVC)$
- $STC \geq TC_{LR}$ for all q

LO Checklist

- LO1: Econ vs. accounting cost; sunk costs ✓
- LO2: Isocost line ✓
- LO3: Cost minimisation ($MRTS = w/v$) ✓
- LO4: Total cost function ✓
- LO5: AC and MC; MC crosses AC at bottom ✓
- LO6: Economies of scale vs. RTS ✓
- LO7: SR cost curves; LRAC envelope ✓

The thread: Opportunity cost $\rightarrow$ isocost line $\rightarrow$ cost minimisation ($MRTS = w/v$) $\rightarrow$ total cost function $\rightarrow$ AC and MC (MC crosses AC at minimum: $dAC/dq = (MC - AC)/q$) $\rightarrow$ economies of scale $\rightarrow$ short-run cost curves $\rightarrow$ LRAC envelope ($SR \geq LR$). Week 7 uses MC to derive the supply curve.

6.11. Key Terms

Opportunity Cost

Value of the best alternative foregone. Foundation of economic cost. *Used throughout the course.*

Accounting Cost

Actual cash payments; appears on financial statements.

Economic Cost

Accounting cost plus all opportunity costs. Excludes sunk costs.

Sunk Cost

A cost already incurred and unrecoverable; irrelevant to forward-looking decisions.

Economic Profit

Total revenue minus economic cost. Can be negative even when accounting profit is positive.

Total Cost (TC)

$wL + vK$; total spending on all inputs. *Used throughout the course.*

Isocost Line

All (L, K) with the same total cost; slope $= -w/v$. Production analogue of the budget line.

Cost Minimisation

Choosing the cheapest input combination to produce a given output; equivalent to profit maximisation for fixed output.

Optimality Condition

$MP_L/MP_K = w/v$ (or $MP_L/w = MP_K/v$); satisfied at the cost-minimising input bundle. *Used throughout the course.*

Bang for the Buck

Rearranged optimality condition: output per dollar from each input must be equal at the optimum.

Average Cost (AC)

TC/q ; cost per unit of output. *Used throughout the course.*

Marginal Cost (MC)

dTC/dq ; slope of the TC curve; cost of the next unit.

MC-AC Identity

$dAC/dq = (MC - AC)/q$; proves MC crosses AC exactly at AC's minimum. Foundation of the grade analogy.

Economies of Scale

AC falling as output rises. Linked to IRS in long run.

Diseconomies of Scale

AC rising as output rises. Linked to DRS in long run.

Efficient Scale

Output that minimises AC; the bottom of the U-shaped AC curve.

Short Run

Period with at least one fixed input; capital fixed in this course.

Long Run

Period when all inputs are variable.

TFC, TVC, STC

Total fixed cost, total variable cost, short-run total cost. $STC = TFC + TVC$.

ATC, AVC, AFC, SMC

Short-run average total/variable/fixed cost and marginal cost. $ATC = AVC + AFC$.

LRAC (Envelope)

Long-run average cost = lower envelope of all SRAC curves. $LRAC \leq SRAC$ everywhere.

Recommended Videos

Four videos selected for clarity and direct relevance to this chapter.

1. Accounting Profit vs. Economic Profit (*Khan Academy*) ~7 min

<https://www.youtube.com/watch?v=kQRFfAlj9CE>

Explains the distinction between explicit and implicit costs, opportunity cost, and why economic profit differs from accounting profit. Directly reinforces Section 6.1 and is ideal before Level 1 questions on cost concepts.

2. Isoquant and Isocost: Cost Minimisation (*Guy Pascale*) ~9 min

<https://www.youtube.com/watch?v=1T8eSU9pxcw>

Visual walkthrough of the isocost line, tangency with the isoquant, and the $MRTS = w/v$ optimality condition. Reinforces Sections 6.2–6.4 and is recommended before Level 2 algebraic cost-minimisation problems.

3. Marginal Cost, AVC, and ATC (*Khan Academy*) ~11 min

<https://www.youtube.com/watch?v=mW6Cwpfb5CI>

Explains why AC is U-shaped, derives the MC–AC relationship, and works through the fixed/variable cost decomposition. Directly mirrors Sections 6.6–6.8. Recommended before the short-run cost questions in Level 2–3.

4. Economies of Scale and Long-Run Costs (*Jacob Clifford, ACDC Economics*) ~10 min

<https://www.youtube.com/watch?v=JdCgu1sOPDo>

Demonstrates how LRAC is derived as the lower envelope of SRAC curves, with the efficient scale and the connection to returns to scale. Reinforces Sections 6.7 and 6.9 and the worked LRAC example.

6.12. Practice Questions

How to use these questions. Difficulty: ★ easy ★★ intermediate ★★★ challenging.
Attempt every question independently before consulting the solutions.

Level 1: Conceptual and Foundational

★ **[Opportunity and Accounting Costs]** A lawyer earns \$150,000/year at a firm. She considers opening her own practice with annual expenses: office rent \$24,000, staff \$60,000, supplies \$6,000.

- (a) Accounting cost of running her own practice?
- (b) Full economic cost including opportunity cost of her time?
- (c) Revenue \$200,000: accounting profit? Economic profit?
- (d) Should she switch? Justify using economic profit.

★ **[Sunk Costs]**

- (a) True or false: “A firm should continue producing as long as revenues cover total costs including sunk costs.” Explain.
- (b) A retailer signed a 5-year lease at \$5,000/month. Two years in, revenues fall sharply. An advisor says: “The lease is gone — focus on whether revenues cover your other costs.” Is she correct? What economic principle applies?

★ **[Isocost Line]** $w = \$6$, $v = \$3$, $TC = \$60$.

- (a) Write the isocost equation and find both intercepts.
- (b) Slope and economic interpretation?
- (c) w rises to \$9: how does the isocost change? Sketch both.
- (d) Instead TC rises to \$90 (original prices): how does the isocost change?

★ **[MC–AC Relationship]** True or false, with explanation:

- (a) When MC equals AC , AC is at its minimum.
- (b) If MC is increasing, AC must also be increasing.
- (c) AC can never exceed MC .
- (d) AFC always decreases as output increases.

★ **[Short Run vs. Long Run]** State short-run or long-run, and which input is fixed:

- (a) Pizza restaurant hires two extra delivery drivers for Friday night.
- (b) Steel mill builds a new smelting furnace (18 months to complete).

- (c) Airline adds flights using existing planes.
- (d) Airline purchases five new aircraft.

★ [Economies of Scale]

- (a) Explain the difference between “returns to scale” and “economies of scale.”
- (b) A firm doubles all inputs and output more than doubles. What does this imply about (i) returns to scale? (ii) long-run economies of scale?
- (c) Can a firm face economies of scale in the short run even with CRS in production? Explain.

★ [Fixed and Variable Costs] $STC(q) = 500 + 4q + 0.5q^2$.

- (a) Identify TFC and TVC.
- (b) Derive AFC, AVC, ATC, SMC.
- (c) Verify $ATC = AFC + AVC$.
- (d) At what output does $SMC = AVC$?

★ [Explain to a Friend — Level 1] *No formulas.* Your friend says: “If a company is making accounting profit, surely it’s doing well — why would an economist say it might be wasting resources?” Write 3–4 plain-English sentences explaining economic profit using a concrete example.

Level 2: Intermediate Analysis

★★ [Cost Minimisation — Algebraic] $q = K^{1/3}L^{2/3}$, input prices w and v .

- (a) Compute MP_L and MP_K .
- (b) Apply the optimality condition to express K in terms of L, w, v .
- (c) Find $L^*(q)$ and $K^*(q)$.
- (d) Derive $TC(q, w, v)$.
- (e) Compute AC and MC. What type of returns to scale? ★ *Week 7: this MC curve is the competitive supply curve.*

★★ [Short-Run Cost Functions] $q = 5L^{1/2}$ (short run, $\bar{K} = 1$), $w = \$20$.

- (a) Express L as a function of q .
- (b) Derive $TVC(q)$.
- (c) With $TFC = \$200$: derive STC, SAC, SMC .
- (d) At what output is SAC minimised?

(e) Show that SMC passes through the minimum of SAC.

★★ [Isocost and Isoquant Analysis] $q = LK$, $w = \$2$, $v = \$8$.

(a) For $TC = \$80$: write the isocost and find its intercepts.

(b) Using the optimality condition, find cost-minimising (L^*, K^*) for $q = 32$.

(c) Minimum cost of $q = 32$?

(d) w rises to $\$8$. New (L^*, K^*) for $q = 32$? New minimum cost? How does the input mix change?

★★ [“Bang for the Buck”] $MP_L = 40$, $MP_K = 20$, $w = \$8$, $v = \$5$.

(a) Compute MP_L/w and MP_K/v .

(b) Is the firm cost-minimising?

(c) In which direction should it reallocate spending?

(d) What condition holds at the new optimum?

★★ [Numerical Short-Run Costs] $STC = 200 + 10q - 4q^2 + q^3$.

(a) Find TFC and TVC.

(b) Find AFC, AVC, ATC, SMC.

(c) Output minimising AVC? Minimum AVC?

(d) Output minimising ATC? Verify $SMC = ATC$ there.

(e) Complete:

q	AVC	ATC	SMC	AFC
1				
2				
3				
4				
5				

★★ [Long-Run vs. Short-Run Costs] $LRTC(q) = 2q$; short-run: $TFC = \$50$, $TVC = q^2/50 + q$.

(a) Find LRAC and LRMC.

(b) Find SAC and SMC.

(c) At what output is $LRTC = STC$?

(d) Show $SAC \geq LRAC$ for all q , with equality at the output in (c).

(e) Sketch both curves.

★★ [Data-Based Question] The table shows a firm's short-run costs.

q	STC	TFC	TVC	Notes
0	50			No output
5	80			
10	130			
15	210			
20	330			
25	500			

- Fill in TFC and TVC. Verify $STC = TFC + TVC$ at each row.
- Compute SMC between consecutive observations. Is SMC U-shaped over this range?
- Compute ATC, AVC, AFC at each observation. At which output is ATC lowest?
- Does SMC cross ATC from below at its minimum? Verify numerically.
- If market price is \$12/unit, at which output should the firm produce?
- ★ *Week 7 preview*: Identify minimum AVC. At what price would the firm shut down in the short run?

★★ [Explain to a Friend — Level 2] Your friend says: "I know MC crosses AC at its minimum, but it seems like a coincidence. Why does this always happen?" Write a short paragraph (4–5 sentences) explaining the logic using the student grade analogy.

Level 3: Advanced Problems

★★★ [Full Cost Derivation: Cobb-Douglas, CRS] $q = L^a K^b$, $a + b = 1$.

- Derive MP_L and MP_K .
- Apply the optimality condition.
- Derive $TC(q, w, v)$.
- Show $MC = AC$ everywhere. What does this imply?
- If $a + b > 1$ (IRS): how does TC shape change? Is AC increasing or decreasing?

★★★ [Multi-Part Cost Analysis] $STC = 0.1q^3 - q^2 + 10q + 50$.

- Derive SMC, ATC, AVC.
- Output minimising SMC?
- Output minimising AVC? Verify $SMC = AVC$ there.

(d) Output minimising ATC? Verify $SMC = ATC$ there.

(e) Sketch ATC, AVC, SMC. Label all three minima.

★★★ [Restaurant Cost Minimisation] $q = 4L^{1/2}K^{1/2}$, $w = \$16$, $v = \$4$.

(a) Derive $L^*(q)$ and $K^*(q)$.

(b) Find long-run $TC(q)$.

(c) Find AC and MC. What does the relationship say about returns to scale?

(d) Short run with $\bar{K} = 25$: (i) express $L(q)$; (ii) find STC, SAC, SMC; (iii) at what q does $STC = TC_{LR}$? ★ At this output SR and LR supply coincide.

★★★ [Economies of Scale and LRAC] $LRAC(q) = q^2 - 8q + 20$.

(a) Find q^* minimising LRAC and the minimum AC.

(b) Range with economies of scale? Diseconomies?

(c) Derive LRMC from LRTC. Show it passes through the LRAC minimum.

(d) Sketch LRAC and LRMC. Label the efficient scale and both regions.

★★★ [Short-Run Constraint vs. Long-Run Optimum] $q = L^{1/2}K^{1/2}$, $w = 4$, $v = 1$.

(a) Long-run: find $L^*(q)$, $K^*(q)$, $LRTC(q)$.

(b) Short run $\bar{K} = 25$: derive STC, SAC, SMC.

(c) At what output does $STC = LRTC$? Verify $SAC = LRAC$ there.

(d) At $q = 10$ and $q = 50$: compute SAC and LRAC. Confirm $SAC \geq LRAC$.

(e) Sketch LRAC and SAC. Mark the tangency point.

★★★ [Conceptual Essay: Cost Concepts]

(a) A company owns a warehouse, paid \$2m ten years ago. Accountant reports zero occupancy cost. An economist disagrees. Who is right?

(b) A firm has run a product line at a loss for two years, citing \$5m in development costs. An advisor recommends shutting it down. What principle supports the advisor?

(c) Why must SRAC lie on or above LRAC at every output? When do they touch?

★★★ [Isocost Changes and Input Substitution]

(a) Why does a wage increase rotate the isocost around the capital-axis intercept rather than shifting it parallel?

(b) Minimum wage rises. Predict the change in K/L at a previously-optimal firm.

Use the bang-for-the-buck condition.

(c) Both w and v double. How does the cost-minimising input combination change? Total cost?

(d) Firm A: $w = \$10$, $v = \$10$. Firm B: $w = \$5$, $v = \$20$ (identical production functions). Which is more labour-intensive? Why?

★★★ **[Explain to a Friend — Level 3]** Your friend says “firms should always minimise their total costs.” Write 5–6 sentences addressing: (a) why this is incomplete; (b) the correct objective; (c) why a firm in the short run may not be at the long-run cost minimum but may rationally continue operating; and (d) a real-world scenario where ignoring the SR/LR distinction leads to a bad decision.

Full solutions follow in the solutions document.